Honours 2021 Introductory Essay

Oriented External Electric Field Catalysis Within Ionic Liquids

Adrea Snow (31883893) - Supervisor: Prof. Ekaterina Pas

1 Introduction

1.1 Oriented External Electric Fields as catalysts

Since chemical reactivity can be broadly reduced down the the dynamics of electrons, if one can control the electronic environment, one can control the reactivity. This notion has been at the forefront of chemical research for as long as electron directing groups have been utilised to manipulate the strength of bonds, however the idea of influencing a reaction from a distance using electric fields (EF) has only been explored in the past century.

The methods to this approach were initially quite crude, involving the use of homogenous ionic species,[1] though through the use of early *ab initio* techniques, the mechanisms were able to be more rigorously and precisely investigated. These computational studies in the early 90s sought to understand solvent effects and more generally the effects of, localised EFs on chemical reactions, such as the Friedel-Crafts[2] and Menshutkin[3] reactions. Further evidence for the direct influence of electrostatic catalysis came from biology, as the idea that within an enzyme, an internal electric field (IEF) generated by charged amino acids within the binding site could effectively be creating a pre-organised environment that would stabilise the substrate's transition state (TS).[4]

Whilst developing the valence bond model of chemical reactivity, Shaik[5] recognised that the definitions of chemical bonding that emerged from the theory indicated that all bonds, irrespective of the nature of their component atoms could be described in a polar context. This inspired Sini et al.[6] to explore

the idea, developing a model of bonding that described all bonds as a superposition of a spin paired, purely covalent interaction and two possible ionic interactions, as shown in Equation (1).[7]

$$\Psi_{A-B} = C_1(\mathrm{A{-}B}) + C_2(\mathrm{A^- B^+}) + C_3(\mathrm{A^+B^-}) \tag{1}$$

These theoretical formalisms laid the groundwork for further computational research into the field, with Shaik et al. continuing to explore the effects of external electric fields (EEF). Initially, they created selectivity between competing nonpolar epoxidation and hydroxylation reactions[8] and later, Meir et al.[9] identified that oriented EEFs (OEEF) could afford endo/exo selectivity of Diels-Alder reactions. This theoretical exploration was put to the test experimentally by Aragonès et al.,[10] who used an OEEF directed from a scanning tunnelling microscope (STM) plate to the tip, to not only control the endo/exo selectivity of the reaction, but also to quantify and correlate the effect of the EF strength. While this research was incredibly valuable as an experimental proof of concept for the progression of the field, the practicality of the method left it unsuitable for any kind of reaction on scale.

With sufficient experimental evidence and theoretical understanding of the effect of OEEFs, Shaik et al. formalised the "governing effects and rules for electric-field effects on chemical reacivity",[7] here on referenced as the Shaik rules. Briefly summarised they are as follows:

Rule i) An OEEF that is oriented along a "bond axis" will polarise the bond, favouring one of the ionic species (Equation (1)), eventually cleaving it heterolytically. As this polarisation occurs, the bond dipole increases, causing it to interact more strongly with the OEEF, stabilising the species in the process.

Rule ii) Every reaction possesses a "reaction axis" which follows the flow of electrons from reactant to product, through the TS.

Rule iii) Applying an OEEF along the "reaction axis" will stabilise the TS by increasing the bond ionicity

(as per rule (i)), and symmetry of the HOMO-LUMO overlap of the transition species allowing, will encourage mixing of the charge-transfer states.

Rule iv) A molecule that either possesses a dipole moment or that is polarisable, will rotate to align with a sufficiently strong OEEF

1.2 Extending the use of OEEFs to Solution

While the STM experiments of Aragonès et al.[10] were conveniently able to fix the reactants in place to ensure that the OEEF passed through the reaction axis, when migrating to an experiment where the species are untethered, this isn't possible. Following rule (iv), however, when allowed to rotate freely the reactant will align itself with the OEEF according to it's dipole moment.[7,11] The amount of stabilising energy that is obtained from the OEEF can be quantified through the relationship $\Delta E = 20\vec{F} \cdot \vec{\mu}\,(\mathrm{kJ{\cdot}mol^{-1}})$ [I]. If the OEEF strength, or the magnitude of the dipole moment is strong enough, the stabilisation energy can overcome the rotational kinetic energy, effectively "tweezing" the molecule in place.[7,12]

While this tweezing effect can be useful for rotation locking individual molecules of interest, it also applies to solvent molecules within the bulk. Since the tweezing effect is dependent on the magnitude of $\vec{F}$, as the electric field increases, so too does the stabilisation energy. When a critical field strength is reached, as per rule (i) and rule (iv), the molecules of the bulk align themselves with the field and with each other in crystalline form, seizing up in a process known as *electrofreezing*.[13] This has been demonstrated through the use of non-polarisable and polarisable molecular dynamics, as shown in Figure 1.1.

[I] where $\vec{F}$ is the OEEF strength (in $\mathrm{V{\cdot}\AA^{-1}}$) and $\vec{\mu}$ is the strength of the dipole (in Debye)

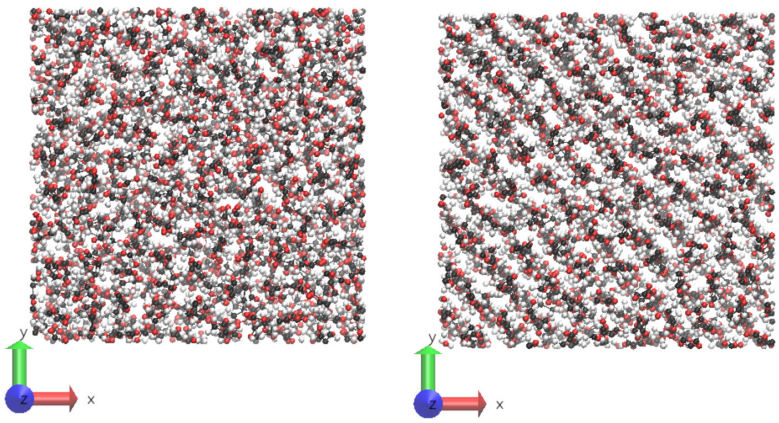

Figure 1.1: Snapshot from a nonpolarisable molecular dynamics simulation of ethylene carbonate in OEEFs of $0.1\,\mathrm{V{\cdot}\AA^{-1}}$ (left) and $0.15\,\mathrm{V{\cdot}\AA^{-1}}$ (right), showing the structure of the solution before and after electrofrezing. Sourced from Daniels et al.[13]

Before this critical field strength is reached, the molecules of the bulk still preferentially align themselves with the OEEF, however they do so to a lesser degree, whilst maintaining their fluidity. This restricted rotational movement does, however result in the solvent molecules being less capable of aligning their dipoles with charged solutes, leading to a significant decrease in dielectric constant.[13] Looking at this phenomena from a different perspective, the OEEF has the ability to break the ion solvation shell,[14] and from the perspective of Marcus theory,[15] presents a reasonable method for reducing the reorganisation energy involved in certain chemical reactions.

While this approach to using OEEFs to catalyse chemical reactions has incredible potential, a potential issue arises; since the OEEF has the ability to catalyse electrochemical reactions through ionising bond cleavage (rule (i)), we need to be sure that the reaction we're performing is occurring via an OEEF stabilised TS (rule (iii)) Xu et al.[11] addressed this issue by devising a method of using the OEEF to preorganise the solvent, but then switching it off, to let the newly formed, solvent induced IEF maintain the field that would catalyse the reaction.[7] This method, however, require the solvent to both be able to maintain the ordered structure from the OEEF and to be able to produce a sufficiently strong IEF.

1.3 Creating IEFs from Ordered Ionic Liquids

Ionic liquids (IL) can be formally described as stoichiometric ratios of complementary ionic species, with a melting point or glass transition temperature[II] of less than $100\,^{\circ}\mathrm{C}$. To maintain their fluidity, the ions chosen are typically bulky organic, phosphate, borate, or sulphate based compounds, which prevent efficient, crystalline packing. As can be seen in Figure 1.2, there is a large amount of diversity in the physical structures that comprise common ILs and as such their physical properties, can be specifically designed for their target application.[17] These include but are not limited to; melting point, dielectric constant, degree of coordination, viscosity, density and conductivity.

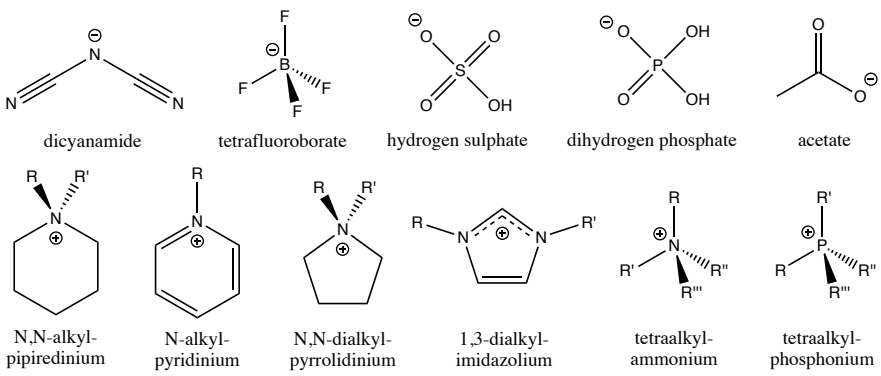

Figure 1.2: A selection of commonly used Ionic liquid ions.

Due to their relative bulk, distributed electronegative components, and/or conjugated nature, the formal charges on ILs are typically delocalised and diffuse which results in them being highly polarisable.[18] Due to rule (iv), this makes them particularly reactive to OEEFs and as such can form and maintain more ordered structures whilst maintaining their fluidity.[19,20] This also means that while their long range intermolecular interactions are naturally predominated by electrostatics, the contribution from Van der Waals forces is significant.[21] These factors combined result in ILs being quite slow to respond to external stimuli as can be seen in their, markedly small self-diffusion coefficients[III].[18]

[II] T_g is used here to exclude substances that are too viscous to be of use at our desired temperature ranges[16]
[III] Experimentally identified as being smaller than those of a conventional solvents by a factor of $\sim 10^3$ [22,23]

Xu et al.[11] recognised both their highly polarisable electron density and their sluggishness as making ILs a good solvent candidate for pulsed OEEF catalysed chemistry. Through computational methods, they demonstrated that utilising a $0.2\,\mathrm{V{\cdot}\AA^{-1}}$ OEEF to order the solvent, an IEF was formed that was able to catalyse an intramolecular proton transfer. This reduced the reaction activation energy by $129.1\,\mathrm{kJ{\cdot}mol^{-1}}$ for [C_2mim][BF_4], compared with the $84.31\,\mathrm{kJ{\cdot}mol^{-1}}$ reduction in the same OEEF using MeOH as the ordered solvent. An assumption was made for both solvents, that the ordered structure and the resulting IEF would be maintained throughout the course of the reaction. While this is understood to likely not be the case for MeOH, further computational studies by the Pas group[IV] suggest that ILs are in fact capable of maintaining their order for significantly longer.

1.4 Addition of H-bonding Networks to IEF forming ILs

A subset of ILs that can be defined are those that are formed from the proton transfer between a Brønsted acid and Brønsted base; these are referred to as protic ionic liquids (PIL).[17] Since the two species within solution will ultimately be in equilibrium with each other, MacFarlane and Seddon[16] proposed that for an PIL to be "pure" it must contain less than 1% neutral species. While the definition of pK_a is limited outside of the context of an aqueous environment, with proton transfer happening orders of magnitude less within pure PILs,[16] a sufficiently large pK_a difference between the cationic and anionic species can be indicative of the PIL's purity.[24]

There is a growing body of evidence that PILs form extensive, extremely strong hydrogen bonding networks, from the works of Swatloski et al.,[25] who explored the powerful ability of imidazolium based ILs to disrupt the hydrogen bonding of cellulose causing it to dissociate, to the work of Brehm et al.[18,26] who explored the dynamics, clustering and formation of [$c_2$mim][OAc] in order to better

[IV] Manuscript in progress

understand the nature of these bonds.

1.5 Computational Chemistry

Computational chemistry is the study of chemistry by means of computational methods. This is often reduced down to the modelling of small, static chemical systems using the equations of quantum mechanics (QM), however one alternate direction in which the the field extends is into the domain of using classical mechanics to model the molecular dynamics (MD) of large chemical systems.

Quantum mechanics (QM) aims to solve the time independent Schrödinger equation ($\hat{H}\Psi = E\Psi$) in order to better understand the nature of the chemical system. *Ab initio* methods accomplish this by optimising a linear combination of primitive gaussian functions, representing the electrons in the system, to form its wavefunction (Ψ). This can then be used with an appropriate hamiltonian ($\hat{H}$) to provide the energy of the system (E). The optimisation of the wavefunction is an iterative process in which each electron's representative functions are optimised against the "mean field" of all other electrons in the system, until the solution reaches the specified convergence criteria. This mean field approach accounts for 99% of the system's energy, however neglects to account for the dynamic components of the motion of the electrons, (correlation energy). When appropriate corrections to this issue are made, these calculations can be highly accurate, however scale with the size of the chemical system in a prohibitively costly manner for anything larger than a handful of atoms.[27]

Rather than optimising the wavefunction of the system directly, density functional theory (DFT) optimises the electron density according to determine the energy of the system. This approach scales far more favourably and more explicitly accounts for the correlation energy than more affordable *ab initio* methods, however the nature of DFT means that each DFT functional is parameterised and thus

exhibits a particular bias based on the training dataset. While this does make the calculations scale more affordably than *ab initio* methods, any deviation beyond what the functional was parameterised for will result in lost accuracy.[27]

In contrast to the approaches taken by QM methods, rather than trying to model one system with perfect accuracy, MD methods use much lower levels of theory to model the dynamics of the system over time, typically utilising the physics of classical mechanics[V] to model a system more approximately. The use of statistical mechanics is then used to determine expectation values from extensive sampling of much longer simulations. This provides multiple benefits, as the simulation can be allowed to *equilibrate* rather than being optimised to an energetic minima, and it allows for the system to evolve dynamically over time. Given that the scaling of these classical equations tends to be significantly more favourable than QM approaches, it also allows for the study of much larger and more complex systems.[27]

1.6 Research Objectives

Following on from the work of Xu et al.,[11] This project aims to investigate the ability of an OEEF to order an IL in order to create and IEF that will catalyse an organic reaction. Their work utilised the BF_4^- anion which has highly non-directional coordination and is incapable of hydrogen bonding. By utilising PILs and tapping into their ability to form extended hydrogen bonding networks, it is hoped that the methodology can be used to catalyse a more complex intramolecular conjugated addition.

Similarly to the work of Xu et al.[11] this will involve the use of DFT based methods to identify the rate determining step (RDS), followed by *ab initio* based methods to parameterise all the species involved to be ordered by the OEEF under MD, which will finally be analysed and compared to the energy of

[V] For example; modelling molecules as highly parameterised charged masses joined together by springs, following the equations of simple harmonic oscillators and Coulombic potentials

the uncatalysed reaction, in order to better understand the processes involved.

While accurate energetics of the catalysed reaction, compared with the uncatalysed reaction would ultimately be ideal, the goals of this research are to gain a better understanding of the behaviour of hydrogen bonded ILs in an OEEF and how they interact with a reacting species.

References

(1) Pocker, Y.; Buchholz, R. F. *Journal of the American Chemical Society* **1970**, *92*, 2075–2084.

(2) Carbonell, E.; Duran, M.; Lledos, A.; Bertran, J. *The Journal of Physical Chemistry* **1991**, *95*, 179–183.

(3) Sola, M.; Lledos, A.; Duran, M.; Bertran, J.; Abboud, J. L. M. *Journal of the American Chemical Society* **1991**, *113*, 2873–2879.

(4) Warshel, A.; Sharma, P. K.; Kato, M.; Xiang, Y.; Liu, H.; Olsson, M. H. M. *Chemical Reviews* **2006**, *106*, 3210–3235.

(5) Shaik, S. S. *Journal of the American Chemical Society* **1981**, *103*, 3692–3701.

(6) Sini, G.; Maitre, P.; Hiberty, P. C.; Shaik, S. S. **1991**, *229*, 163–188.

(7) Shaik, S.; Danovich, D.; Joy, J.; Wang, Z.; Stuyver, T. *Journal of the American Chemical Society* **2020**, *142*, 12551–12562.

(8) Shaik, S.; de Visser, S. P.; Kumar, D. *Journal of the American Chemical Society* **2004**, *126*, 11746–11749.

(9) Meir, R.; Chen, H.; Lai, W.; Shaik, S. *ChemPhysChem* **2010**, *11*, 301–310.

(10) Aragonès, A. C.; Haworth, N. L.; Darwish, N.; Ciampi, S.; Bloomfield, N. J.; Wallace, G. G.; Diez-Perez, I.; Coote, M. L. **2016**, *531*, 88–91.

(11) Xu, L.; Izgorodina, E. I.; Coote, M. L. *Journal of the American Chemical Society* **2020**, *142*, 12826–12833.

(12) Wang, C.; Danovich, D.; Chen, H.; Shaik, S. *Journal of the American Chemical Society* **2019**, *141*, 7122–7136.

(13) Daniels, I. N.; Wang, Z.; Laird, B. B. *The Journal of Physical Chemistry C* **2017**, *121*, 1025–1031.

(14) Novák, M.; Foroutan-Nejad, C.; Marek, R. *Physical Chemistry Chemical Physics* **2016**, *18*, 30754–30760.

(15) Landau, L. *Physical Review* **1949**, *75*, 884–885.

(16) MacFarlane, D. R.; Seddon, K. R. *Australian Journal of Chemistry* **2007**, *60*, 3.

(17) Greaves, T. L.; Drummond, C. J. *Chemical Reviews* **2008**, *108*, 206–237.

(18) Brehm, M.; Weber, H.; Pensado, A. S.; Stark, A.; Kirchner, B. *Physical Chemistry Chemical Physics* **2012**, *14*, 5030.

(19) Wang, Y. *Journal of Physical Chemistry B* **2009**, *113*, 11058–11060.

(20) Zhao, Y.; Dong, K.; Liu, X.; Zhang, S.; Zhu, J.; Wang, J. *Molecular Simulation* **2012**, *38*, 172–178.

(21) Izgorodina, E.; Golze, D.; Maganti, R.; Armel, V.; Taige, M.; Schubert, T. J. S.; MacFarlane, D. R. *Phys. Chem. Chem. Phys.* **2014**, *16*, 7209–7221.

(22) Yoshida, K.; Matubayasi, N.; Nakahara, M. *The Journal of Chemical Physics* **2008**, *129*, 214501.

(23) Harris, K. R.; Kanakubo, M. *Journal of Chemical & Engineering Data* **2016**, *61*, 2399–2411.

(24) Yoshizawa, M.; Xu, W.; Angell, C. A. *Journal of the American Chemical Society* **2003**, *125*, 15411–15419.

(25) Swatloski, R. P.; Spear, S. K.; Holbrey, J. D.; Rogers, R. D. **2002**, 4974–4975.

(26) Brehm, M.; Weber, H.; Pensado, A. S.; Stark, A.; Kirchner, B. *Zeitschrift für Physikalische Chemie* **2013**, *227*, 177–204.

(27) Cramer, C. J., *Essentials of computational chemistry : theories and models*, 2nd ed.; John Wiley & Sons Ltd: 2004, p 596.